

The background of the slide features a complex, futuristic design. On the left, there are concentric circles and gear-like patterns in a lighter blue shade. A small cluster of three squares (two light blue, one dark blue) is positioned above the main text. The right side of the slide is a solid, darker blue. The text is centered in a white, sans-serif font.

Intel® RealSense™ Updated 2024

RealSense™: Industry Leading Computer Vision

Intel RealSense Product Portfolio

Depth Camera Peripherals and Modules

Indoor or outdoor and able to operate multiple cameras in the same scene. Peripheral or embedded modules and ASICs



D435/D435i

.3m-3m*

Z accuracy 2% at 2m
D435i has IMU



D435f/D435if

.3m-3m*

Z accuracy 2% at 2m
IR Pass Filter



D415

.5m – 3m*
Z accuracy 2% at 2m



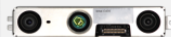
D455/D455f/D456

.6m – 6m*

Z accuracy 2% at 4m
IMU



D450



D430



D415



D410



D4 ASIC Boards



D4 ASIC

GMSL FAKRA

High quality depth for Robotics with Automotive style interface and cables



D457

.6m-6m (Ideal Range)
Uses D450 Module
with GMSL FAKRA Interface
IP65 rated



D450

Short Range

Close-in, wrist-mountable small stereo camera



D405

7cm-50cm*



D401

Tracking w/ IMUs

Depth cameras with IMUs for 3rd Party VSLAM SW



D435i

.3m-3m*

Z accuracy 2% at 2m
IMU



D455/D457

.6m – 6m*

Z accuracy 2% at 4m
IMU

Software from Intel

Intel® RealSense™ SDK 2.0 and sample libraries
(Compatible with Intel® Distribution of OpenVINO™ toolkit)

Intel's RealSense Website: <https://www.intelrealsense.com/>

3D Sensing Brings Value to Variety of Market Segments



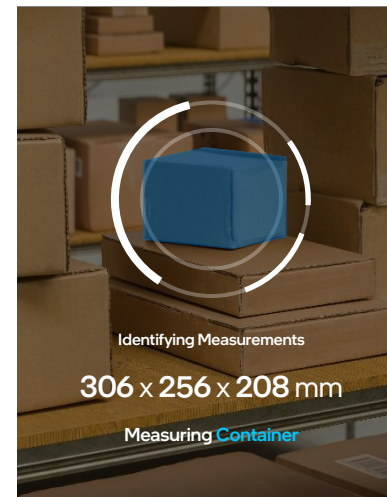
Facial
Authentication



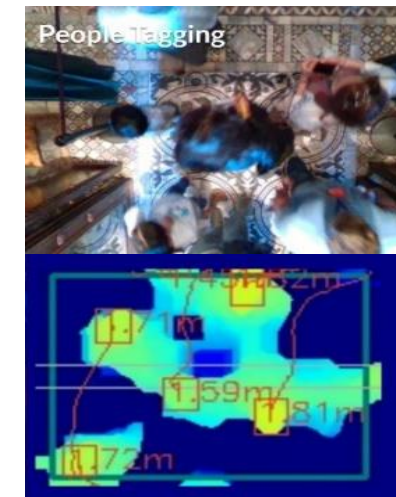
Robotics



Scanning



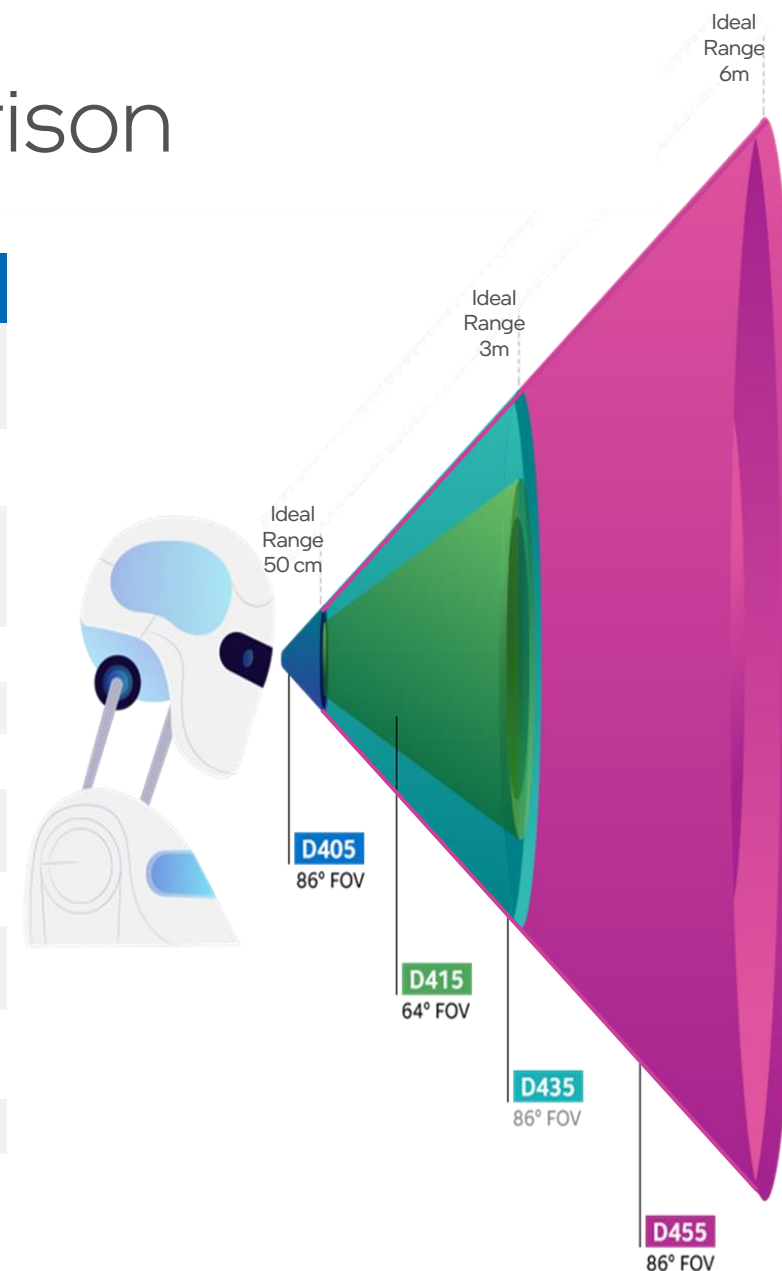
Measurement
and Logistics



Recognition
and Interaction

Intel® RealSense™ Depth Family Comparison

	D405*	D415	D435/D435i/435f	D45x*
Typical Use Case	Pick and Place Defect Detection	Collision avoidance Recognition and Interaction Scanning	Collision avoidance Recognition and Interaction Scanning	Collision avoidance Recognition and Interaction Scanning
Value	Shortest stereo range camera	Best Priced Stereo Camera	WFOV Stereo Camera with or without IMU	Longer Range Stereo Camera
Ideal Range (Stereo cameras can see further but > Distance = accuracy worsens geometrically)	7cm to 50cm	.5m to 3m	.3m to 3m	.6m to 6m
Depth Accuracy²	1.4% at 20cm	2% at 2m	2% at 2m	2% at 4m
Min Z @ Max Resolution	Sub-mm @ 7cm	Min Z ~45cm	Min Z ~28cm	Min Z ~52cm
Environment	Indoor & Outdoor	Indoor & Outdoor	Indoor & Outdoor	Indoor & Outdoor
Max Depth Resolution	1280x720 @ 30fps 640x360 @ 90fps	1280x720 @ 30fps 848x480 @ 90fps	1280x720 @ 30fps 848x480 @ 90fps	1280x720 @ 30fps 848x480 @ 90fps
Depth FOV HD/Shutter¹	87°x58°/Global	65°x40°/Rolling	87°x58°/Global	87°x58°/Global
RGB/Shutter¹	1MP/87°x58°/Global via Left Depth Imager	2MP/69°x42°/Rolling	2MP/69°x42°/Rolling	1MP/90°x65°/Global
Product Dimensions (WxHxD mm)	42x42x22	99x23x20	90x25x25	124x29x26 (D455)
IMU	NO	NO	NO (D435i YES)	YES



¹FOV (HxV) is measured up to +/-3° of stated value; ²Z-accuracy as measured as out of the factory; ³9m at 10% ROI with 95% target reflectivity; * Final technical specs not available until QS for each product